

# Nordita PhD School “Amplitudes, Strong Gravity & Resummation”

## Numerical Relativity Hands-On Exercises

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### Problem 1

- (a) Show that the normalised one-form  $\omega_a$  is given by

$$\omega_a = \alpha \Omega_a. \quad (1)$$

- (b) Show that the covariant derivative  $D$  induced on a spatial hypersurface  $\Sigma$  is compatible with the spatial metric  $\gamma_{ab}$ , i.e.,

$$D_c \gamma_{ab} = 0. \quad (2)$$

- (c) Show that

$$\mathcal{L}_n \gamma_{ab} = -2K_{ab}. \quad (3)$$

Hint: Use the covariant definition of the Lie derivative and the projection operator  $\gamma_a^b = \delta_a^b + n_a n^b$ .

### Problem 2

- (a) Derive the Hamiltonian constraint, i.e.,  ${}^{(3)}R + K^2 - K^{ab}K_{ab} = 16\pi\rho$ , by contracting the Gauss equation twice with the spatial metric, where the Gauss equation is given by:

$$\gamma_a^p \gamma_b^q \gamma_c^r \gamma_d^s {}^{(4)}R_{pqrs} = {}^{(3)}R_{abcd} + K_{ac}K_{bd} - K_{ad}K_{bc}. \quad (4)$$

- (b) Derive the momentum constraint, i.e.,  $D_b K_a^b - D_a K = 8\pi S_a$ , by contraction the Codazzi equation, which is given by

$$\gamma_a^p \gamma_b^q \gamma_c^r n^s {}^{(4)}R_{pqrs} = D_b K_{ac} - D_a K_{bc} \quad (5)$$

Hint: Use  $G_{ab} = {}^{(4)}R - \frac{1}{2}{}^{(4)}R g_{ab} = 8\pi T_{ab}$ .

- (c) Derive the time evolution equations for the extrinsic curvature tensor and the spatial metric, i.e.  $\mathcal{L}_t K_{ab}$  and  $\mathcal{L}_t \gamma_{ab}$ , without making use of the foliation-adapted coordinates but rather by making use of the general definition of the time coordinate,  $t^a = \alpha n^a + \beta^a$ , and the linearity of the Lie derivative, i.e.  $\mathcal{L}_t \rightarrow \mathcal{L}_{\alpha n} + \mathcal{L}_\beta$ .

### Problem 3

Given the Schwarzschild metric in isotropic coordinates,

$$ds^2 = -\frac{(1 - \frac{m}{2r})^2}{(1 + \frac{m}{2r})^2} dt^2 + \left(1 + \frac{m}{2r}\right)^2 (dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2), \quad (6)$$

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consider the foliation given by spacelike hypersurfaces with constant time coordinate  $t$ . Compute the covector  $\Omega_a$ , the lapse  $\alpha$  of this foliation, the unit normal  $n^a$  and the spatial metric  $\gamma_{ab}$ . What are the extrinsic curvature  $K_{ab}$  and the mean curvature  $K$  of this foliation? Is this foliation maximally sliced?

Note: Indices are dummy indices indicating tensorial notation. Recall that the extrinsic curvature and the spatial metric are purely spatial objects. We use the Einstein summation notation, i.e., repeated co- and contravariant indices are summed over.

Recommended literature:

1. T. Baumgarte and S. Shapiro: "Numerical Relativity" (Cambridge University Press), Chapter 2
2. M. Alcubierre: "Introduction to 3 + 1 Numerical Relativity" (Oxford University Press)